

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EE) Examination****March-April-2018****EE 203.02 Electrical Measurement and Measuring Instruments****Date: 02.04.2018, Monday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** Draw the equivalent circuit and phasor diagram for current transformer. **[07]**
- Q – 2.a** Describe the Varley loop test for localization of ground and short circuit faults in cables. **[04]**
- Q – 2.b** **Answer any two questions.** **[10]**
- a) Derive the general equations for balance for an a.c. bridge.
 - b) Explain construction and working of single phase moving iron power factor meter.
 - c) Derive the equation of Maxwell's Bridge for measurement of unknown inductance with neat and clean diagram.
- Q - 3** **Answer any two.** **[14]**
- a) Explain detailed construction and working of single phase induction type energy meter.
 - b) Describe the construction and working of PMMC instrument.
 - c) Describe the working of rectifier type instrument.

SECTION – II

- Q – 4.a** Define the following term **[04]**
1) Accuracy 2) Sensitivity 3) True Value 4) Precision
- Q – 4.b** Explain the difference between limiting and Known errors **[03]**
- Q – 5.a** Describe the basic principle of operation of a d.c. potentiometer. **[05]**

OR

- Q – 5.a** Explain the principle of working of Kelvin's Double Bridge. **[05]**
- Q – 5.b** Draw the Circuit of a Wheatstone bridge and derive the condition of balance. **[04]**
- Q – 5.c** Explain in detail Ammeter-Voltmeter method for measurement of medium order resistance. **[05]**

OR

- Q – 5.c** Describe with the help of suitable diagrams, how a d.c. potentiometer can be used for **[05]**

calibration of a voltmeter.

- Q – 6.a** Explain the any one application of instrument transformers [07]
- Q – 6.b** Explain the method of Reversals for experimental determination of hysteresis loop of magnetic specimen. [07]

OR

- Q – 6.a** Derive the equations for balance in the case of De Sauty's bridge. [07]
- Q – 6.b** Describe the mutual inductance method for testing of a potential transformer. [07]
